HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and optimized **deep learning models** for neural data processing, achieving a **15% improvement** in decoding accuracy on benchmark datasets.
- Collaborated with **research scientists** to advance **generative modeling** initiatives, contributing to hypothesis formulation and experimental design.
- Engineered and maintained **machine learning pipelines** on **AWS**, ensuring high performance and reproducibility across distributed GPU systems for model training.
- Contributed to technical documentation and experimentation for a research publication on **AI-driven neuroscience**, focusing on model evaluation and interpretation.
- Implemented **Python scripts** for data augmentation and feature engineering on **unstructured neural datasets**, processing over **5TB of raw data**.
- Maintained high standards of **code quality** and **reproducibility** by implementing comprehensive unit tests and CI/CD workflows using **Jenkins** and **Git**.
- Identified and resolved performance bottlenecks in **model inference**, reducing p95 latency by **20%** through code optimization and efficient data handling.

SKILLS

ML Frameworks – PyTorch, Hugging Face Transformers, Diffusers, Accelerator, scikit-learn

Programming Languages – Python, SQL, R

Cloud & Infrastructure – AWS (EC2, S3, Lambda, CloudWatch), Docker, Kubernetes, GPU Systems

Data Handling – Large Datasets, Unstructured Data, Data Warehousing, ETL, pandas, NumPy

Methodologies – Deep Learning, Generative AI, Supervised Learning, Self-supervised Learning, Semi-supervised Learning, Representation Learning, Neuroscience Data Analysis, Scientific Experimentation, Agile/Scrum

Tools & Technologies – FastAPI, Jenkins, Git, Jira, Confluence, Linux

PROJECTS

Deep Learning for Neural Signal Decoding

- Developed a **PyTorch**-based deep learning model to decode complex neural signals from simulated EEG data, achieving an **AUC of 0.89** on a held-out test set.
- Engineered a **Python** data processing pipeline using **pandas** and **NumPy** to clean and transform raw neural time-series data, handling approximately **2TB of input**.
- Implemented model evaluation metrics and conducted rigorous scientific experimentation to compare different architectures, documenting findings in a technical report.

Generative AI for Synthetic Neuroscience Data

- Built a **generative model** using **Hugging Face Diffusers** to create synthetic neural activity patterns, supplementing real datasets for model training.
- Integrated the generative model into an existing **Python** inference pipeline, enabling the creation of over **10,000 synthetic samples** per hour.
- Collaborated with a senior researcher to define requirements for synthetic data realism and utility in downstream machine learning tasks.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)